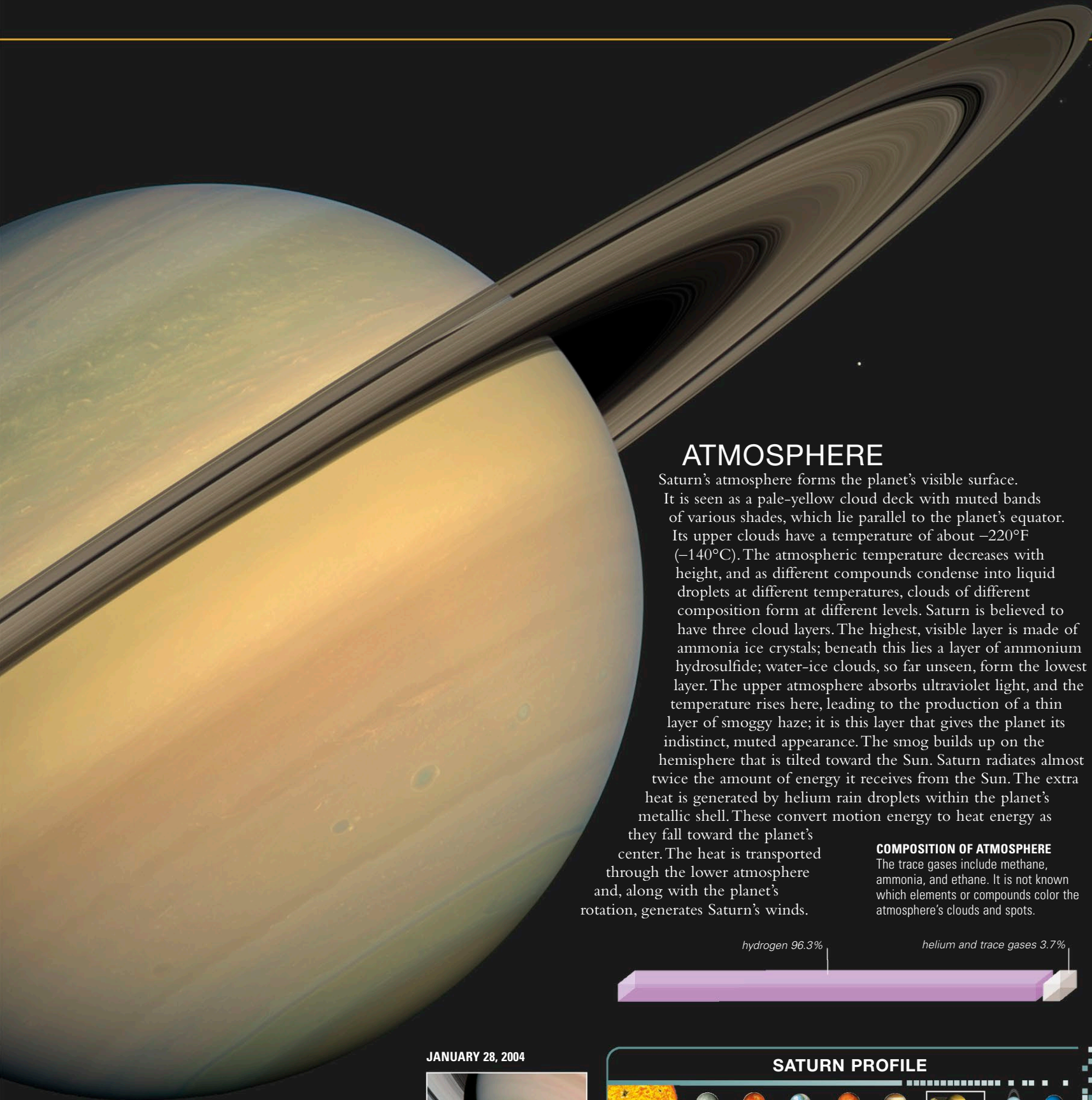




ATMOSPHERE

Saturn's atmosphere forms the planet's visible surface. It is seen as a pale-yellow cloud deck with muted bands of various shades, which lie parallel to the planet's equator. Its upper clouds have a temperature of about -220°F (-140°C). The atmospheric temperature decreases with height, and as different compounds condense into liquid droplets at different temperatures, clouds of different composition form at different levels. Saturn is believed to have three cloud layers. The highest, visible layer is made of ammonia ice crystals; beneath this lies a layer of ammonium hydrosulfide; water-ice clouds, so far unseen, form the lowest layer. The upper atmosphere absorbs ultraviolet light, and the temperature rises here, leading to the production of a thin layer of smoggy haze; it is this layer that gives the planet its indistinct, muted appearance. The smog builds up on the hemisphere that is tilted toward the Sun. Saturn radiates almost twice the amount of energy it receives from the Sun. The extra heat is generated by helium rain droplets within the planet's metallic shell. These convert motion energy to heat energy as they fall toward the planet's center. The heat is transported through the lower atmosphere and, along with the planet's rotation, generates Saturn's winds.

COMPOSITION OF ATMOSPHERE

The trace gases include methane, ammonia, and ethane. It is not known which elements or compounds color the atmosphere's clouds and spots.

hydrogen 96.3%

helium and trace gases 3.7%



JANUARY 28, 2004

JANUARY 26, 2004

JANUARY 24, 2004



CHANGING SOUTH POLAR AURORA

Solar-wind particles in the upper atmosphere produce aurorae. The brightening of the aurora on January 28 corresponds with the arrival at Saturn of a disturbance in the solar wind.

SATURN PROFILE

AVERAGE DISTANCE FROM THE SUN 888 million miles (1.43 billion km)	ROTATION PERIOD 10.66 hours
CLOUDTOP TEMPERATURE -220°F (-140°C)	ORBITAL PERIOD (LENGTH OF YEAR) 29.46 Earth years
DIAMETER 74,898 miles (120,536 km)	MASS (EARTH = 1) 95
VOLUME (EARTH = 1) 763.59	GRAVITY AT CLOUDTOPS (EARTH = 1) 1.07
NUMBER OF MOONS 82	SIZE COMPARISON
OBSERVATION Saturn is visible to the naked eye for about 10 months of the year. It appears like a star and takes about 2.5 years to pass through one zodiacal constellation. A telescope is needed to make out the ring system.	EARTH SATURN